

Regular Operations

The class of regular languages is closed under union or

Regular Operations $\cup, *, \circ$

$A \cup B \Rightarrow$

Regular

Regular

$A \cup B = A \cup B$

$A \circ B = A \circ B$

Start $\rightarrow$



$\{ \text{Starts with } 111 \}$ $\rightarrow$ $A \cup B$ $\rightarrow$ $\{ \text{Ends with } 0 \}$

With all $\cup, *, \circ = \sum_i (\text{Alphabet})$

You can create all Regular Languages

Regular languages are closed under regular operations

From Previous Lessons

CONCATENATION ($\circ$)

$A = \{ \text{mango, orange} \}$

$B = \{ \text{green, orange} \}$

orangeorange

$A \circ B = \{ \text{mangogreen, mangoorange, orangegreen, orangeorange} \}$

KLEENE STAR ($*$)

Concatenate together strings in the language ZERO or more times

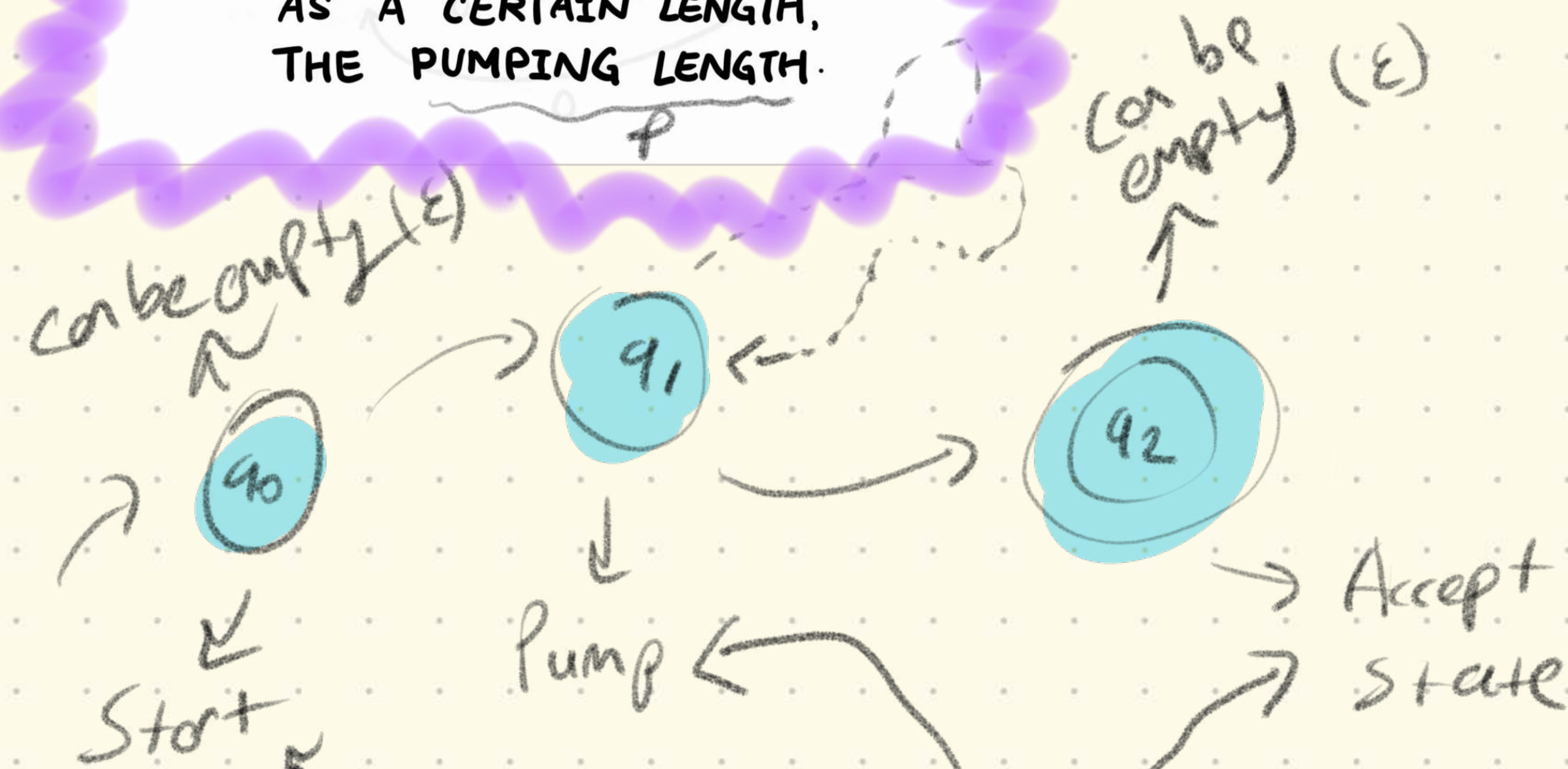
$A = \{ \text{mango, orange} \}$

$A^* = \{ \text{orange, mango, orangeorange, orangemango, mangoorange, mangomango, orangeorangeorange, orangeorange mango, orangemangoorange} \}$

Pumping Lemma

PUMPING LEMMA:

ALL STRINGS IN THE LANGUAGE
CAN BE REPEATED—"PUMPED"—
IF THEY ARE AT LEAST AS LONG
AS A CERTAIN LENGTH,
THE PUMPING LENGTH.



1. for each $i \geq 0$, $xy^iz \in A$
2. $|y| > 0 \rightarrow$ CANNOT BE EMPTY (ϵ)
3. $|xy| \leq p$ $\rightarrow$ pumping length

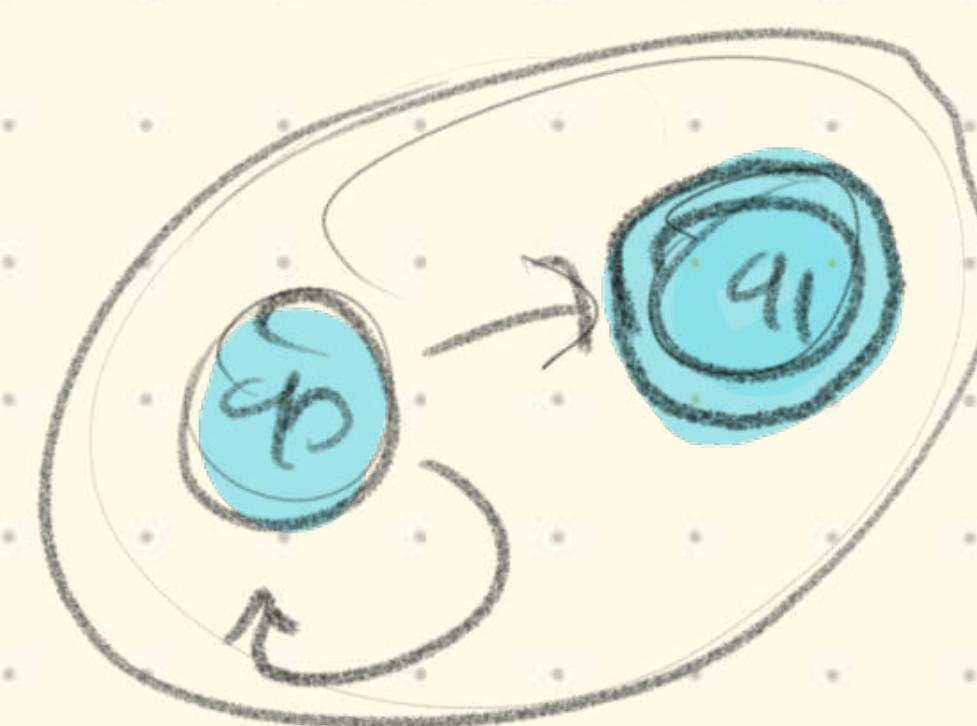
if $p=3 \rightarrow$

must be within
first $|xy| \leq 3$ $\checkmark$

01111

This "111"
is not in the
area between
start $\leftarrow x$ and $y \rightarrow$ Pump

We know that Regular Languages can
be expressed by



Finite
Automaton

OR

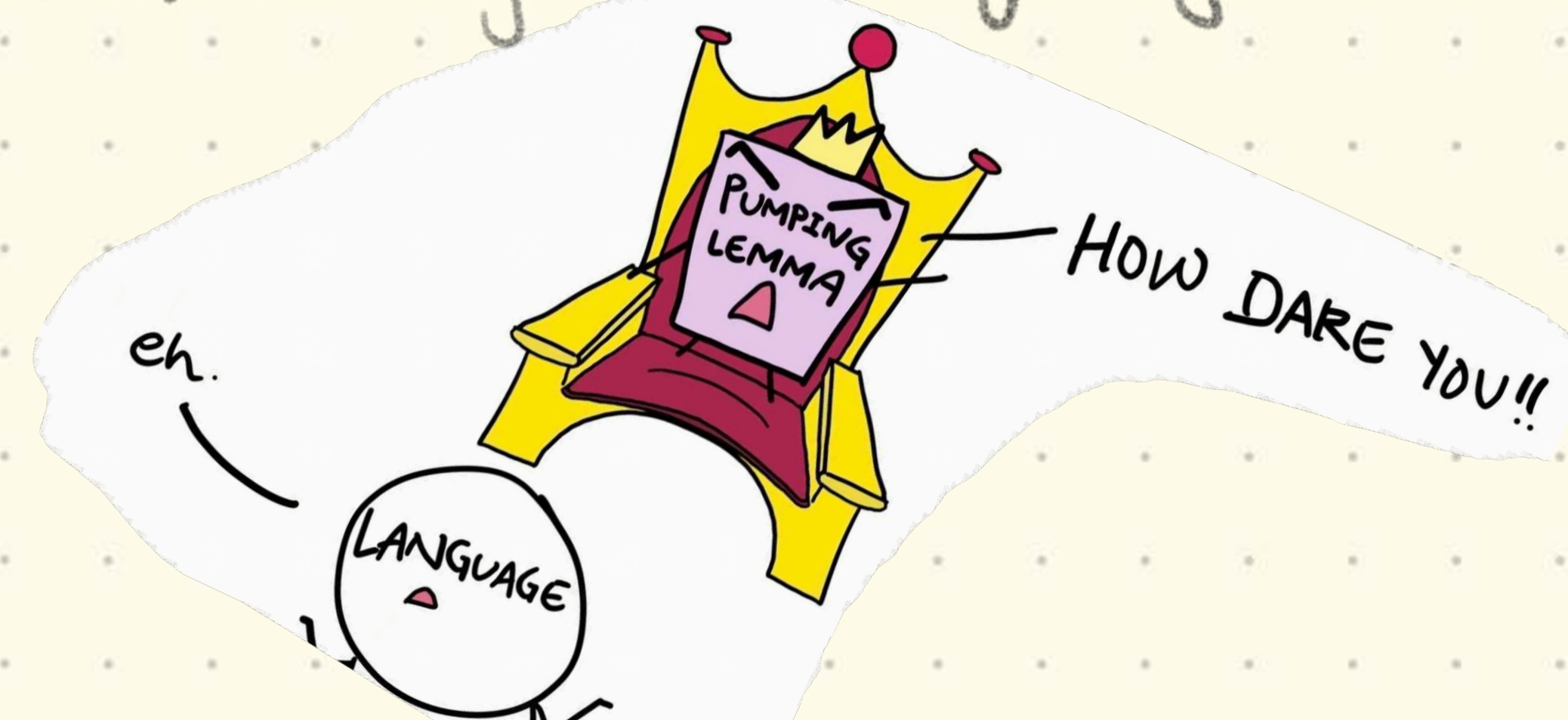
$\{011\}^* \{1011\}^*$
Regular expression

We also know closure properties

$A \cup B$, $A \cap B$, $A \circ B$, A^*

Proof By Contradiction

is for satisfying the pumping lemma
• Regular Languages satisfies $p.l.$
but Non-regular languages not.



Exp $B = \{0^n 1^n \mid n \geq 0\}$
Is this language regular or non-regular?

Try yourself

Answer Let's see if this language satisfies the pumping lemma
so we have $x y^i z^{(i>0)}$ rule in p.l.
 $|y| > 0$ and $|xy| \leq p$

- ① $\underbrace{0^p}_{y} 1^p \Rightarrow 00000 \cdot 0^p 1^p \notin B$ (num of 0 $\downarrow$ num of 1)
- ② $\underbrace{0^p 1^p}_{y} \Rightarrow 0^p 1^p 1111 \rightarrow \notin B$ (num of 1 $>$ num of 0)
- ③ $\underbrace{0^p 1^p}_{y} \Rightarrow 0 \underbrace{00101010101}_{y} \dots 1 \notin B$ (language is not regular)

so this language doesn't SATISFY } Also if we apply the $|xy| \leq p$ rule, we can eliminate ② and ③ because

The pump wasn't at the start of the regular expression. Which has to be at the start.